

Computer Vision Prepathon — Technical Report

1. Introduction

This project addresses two computer vision problems from the Computer Vision Prepathon:

- **Task A:** Automated segmentation of solar filaments in full-disk H-alpha images.
- **Task B:** Diagnosing and reducing object hallucination in Vision-Language Models (VLMs).

Task A focuses on accurately identifying solar filament pixels, including thin and difficult structures. Task B investigates whether a lightweight confidence-based decision strategy can reduce false-positive object hallucinations in a VLM without requiring full model fine-tuning.

For Task A, a U-Net segmentation model was developed as a baseline and then enhanced with an **edge-aware loss** to improve the handling of filament boundaries.

For Task B, the **BLIP VQA model** was evaluated on the POPE hallucination benchmark. Three approaches were compared: the original baseline, naive prompting, and a confidence-based decision strategy.

2. Task A — Automated Segmentation of Solar Filaments

2.1 Objective

The objective of Task A is to perform pixel-level segmentation of solar filaments in full-disk H-alpha images.

The main challenge is that solar filaments can be very thin, low-contrast, and irregularly shaped. The segmentation model therefore needs to identify the filament material while preserving its structure and boundaries.

The task evaluation includes metrics such as IoU, Precision, Recall, AP@IoU, Hit Rate, Miss Rate, and Multi-scale IoU.

2.2 Dataset

The MAGFiLO dataset was used for the experiment.

The dataset contains manually annotated solar filaments from full-disk H-alpha observations. The downloaded dataset contained:

- **704 solar images**
- **1,143 image entries in the annotation file**
- **8,212 annotations**
- Polygon-based filament annotations

The polygon annotations were converted into binary segmentation masks. When multiple filament annotations were present in an image, they were combined into a single semantic mask.

A total of **704 image-mask pairs** were successfully generated.

The dataset was divided into:

- **80% training data:** 563 images
 - **20% validation data:** 141 images
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2.3 Preprocessing

The solar images were converted to grayscale and resized to **256 × 256 pixels**.

The masks were also resized to 256 × 256 using nearest-neighbor interpolation to preserve the binary mask structure.

The input pixel values were normalized to the range:

[0,1][0,1]

The processed images and masks were loaded using PyTorch DataLoaders with a batch size of 8.

2.4 Baseline U-Net

A standard U-Net architecture was used as the baseline segmentation model.

U-Net was selected because it is widely used for image segmentation and combines:

- An encoder for extracting image features
- A bottleneck for learning high-level representations
- A decoder for reconstructing the segmentation
- Skip connections for preserving spatial information

The model was trained using a combination of:

- Binary Cross Entropy loss
- Dice loss

The optimizer was Adam with a learning rate of:

The baseline model was trained for **20 epochs**.

2.5 Edge-Aware U-Net

A major difficulty in solar filament segmentation is preserving thin structures and accurate boundaries.

To address this, an **edge-aware loss** was introduced.

Sobel filters were used to extract horizontal and vertical edges from both the predicted mask and the ground-truth mask.

The additional edge loss encourages the predicted segmentation to have boundaries similar to the ground-truth segmentation.

The final loss was:

$$L = L_{\text{BCE}} + L_{\text{Dice}} + 0.1L_{\text{Edge}} = L_{\{\text{BCE}\}} + L_{\{\text{Dice}\}} + 0.1L_{\{\text{Edge}\}}$$

This provides a lightweight modification to the baseline model without requiring a completely different architecture.

2.6 Task A Results

The main pixel-level results are:

Metric	Baseline U-Net	Edge-Aware U-Net
IoU	0.4602	0.4671
Precision	0.5930	0.6228
Recall	0.6726	0.6513

The edge-aware model produced a higher IoU and substantially improved precision.

The recall decreased slightly, showing a trade-off between detecting more filament pixels and producing cleaner predictions.

2.7 Instance-Level Evaluation

Because the implemented model is a semantic U-Net rather than an instance-segmentation model, instance AP was estimated by:

1. Treating the annotated polygons as ground-truth instances.
2. Extracting connected components from the predicted segmentation.
3. Using the mean predicted probability as the confidence score.
4. Matching predicted components to ground-truth instances using IoU.

At a prediction threshold of **0.3** and IoU threshold of **0.5**, the results were:

Metric	Result
AP@IoU 0.5	0.2449
Precision	0.3425
Recall	0.4472
True Positives	525
False Positives	1008
False Negatives	649

The corresponding:

Hit Rate=0.4472Hit\ Rate = 0.4472

and

Miss Rate=0.5528Miss\ Rate = 0.5528

2.8 Multi-Scale Evaluation

The model was also evaluated at multiple image scales.

Scale	IoU
1.00	0.4667
0.75	0.3468
0.50	0.3486
0.25	0.1694

The resulting Multi-scale IoU was:

MIoU=0.3329MIoU = 0.3329

The decrease in performance at smaller scales indicates that very thin filament structures are difficult to preserve when the image resolution is reduced.

2.9 Visual Results

The segmentation results demonstrate the differences between the ground-truth masks, baseline U-Net predictions, and edge-aware U-Net predictions.

The uploaded result images contain:

Original Image → **Ground Truth** → **Baseline U-Net** → **Edge-Aware U-Net**

These visualizations provide qualitative evidence of the segmentation performance and show how the edge-aware approach changes the predicted filament boundaries.

2.10 Failure Analysis

Failure cases were examined by identifying images with very low segmentation IoU.

Some of the worst cases had:

- Very thin filament structures
- Small annotated regions
- Low contrast
- Difficult boundaries

In these cases, the model sometimes produced almost-empty predictions even when filament structures were present in the ground truth.

This indicates that small and low-contrast structures remain a major challenge.

2.11 Task A Novelty

The main novelty introduced in Task A is the use of an **edge-aware loss function** together with a standard U-Net.

Instead of relying only on pixel-wise BCE and region-overlap Dice loss, the additional Sobel-based edge loss explicitly encourages the model to preserve filament boundaries.

The approach is lightweight and does not require changing the complete U-Net architecture.

3. Task B — Diagnosing Object Hallucination in Vision-Language Models

3.1 Objective

The objective of Task B is to investigate object hallucination in Vision-Language Models.

Object hallucination occurs when a model predicts that an object is present in an image even though it is not actually present.

The experiment investigates whether a lightweight confidence-based strategy can reduce these false-positive predictions.

3.2 Dataset

The **POPE (Polling-based Object Probing Evaluation)** dataset was used.

The dataset contains:

- **9,000 questions**
- 3,000 adversarial samples
- 3,000 popular samples
- 3,000 random samples

The questions require a binary **yes/no** answer about whether an object is present in an image.

For the main experiments, a subset of **1,000 samples** was evaluated because of computational constraints.

Category-wise evaluation was performed using **300 samples from each category**.

Therefore, the reported experimental results should not be interpreted as a full 9,000-sample evaluation.

3.3 Vision-Language Model

The model used was:

Salesforce BLIP VQA Base

BLIP was selected because it provides a pretrained vision-language question-answering model capable of answering questions about images.

The model was run using PyTorch on a GPU.

4. Task B Experiment 1 — Baseline

The original BLIP model was evaluated without additional hallucination-specific prompting or calibration.

The baseline results on the evaluated 1,000 samples were:

Metric	Baseline
Accuracy	82.30%
Precision	84.58%
Recall	79.00%
F1	81.70%
False Positive Rate	14.40%
Yes Ratio	46.70%

The confusion matrix was:

	Predicted No	Predicted Yes
Actual No	428	72
Actual Yes	105	395

The baseline produced **72 false positives**, which represents the main hallucination problem investigated in this task.

5. Task B Experiment 2 — Naive Prompting

A simple prompt was introduced:

“Answer only yes or no. Carefully inspect the image. Say yes only if the object is clearly visible. If you are uncertain, answer no.”

The intention was to encourage the model to inspect the visual information more carefully.

However, the results showed that naive prompting actually made performance worse.

Metric	Baseline	Naive Prompt
Accuracy	82.30%	67.77%
Precision	84.58%	62.90%
Recall	79.00%	86.80%
F1	81.70%	72.94%

FPR 14.40% **51.30%**

The false-positive rate increased dramatically from:

14.4%→51.3%14.4\% \rightarrow 51.3\%

This experiment demonstrates that simply telling a VLM to “carefully inspect” the image does not necessarily improve visual grounding.

Instead, the prompt caused the model to produce many more positive answers.

This failed experiment was therefore useful because it demonstrated that prompt-only intervention can introduce an undesirable decision bias.

6. Task B Experiment 3 — Confidence-Based Mitigation

A lightweight confidence-based strategy was then developed.

Instead of directly using the model's generated yes/no answer, the probability of the first generated “**yes**” and “**no**” tokens was extracted.

A normalized probability for “yes” was calculated as:

$$P(\text{yes}) = \frac{P_{\text{yes}}}{P_{\text{yes}} + P_{\text{no}}}$$

Different confidence thresholds were then tested.

Threshold	Accuracy	Precision	Recall	F1	FPR
0.30	82.10%	80.57%	84.60%	82.54%	20.40%
0.40	82.80%	83.61%	81.60%	82.59%	16.00%
0.50	82.30%	84.58%	79.00%	81.70%	14.40%
0.60	83.20%	87.39%	77.60%	82.20%	11.20%
0.70	83.30%	88.99%	76.00%	81.98%	9.40%
0.80	82.50%	90.73%	72.40%	80.53%	7.40%

A threshold of **0.70** was selected as the final operating point.

The 0.80 threshold produced an even lower FPR, but it reduced recall and F1 more significantly.

7. Final Task B Results

At the selected confidence threshold of 0.70:

Metric	Baseline	Confidence-Based
Accuracy	82.30%	83.30%
Precision	84.58%	88.99%
Recall	79.00%	76.00%
F1	81.70%	81.98%
FPR	14.40%	9.40%

The confusion matrix changed from:

Baseline:
[[428, 72],
[105, 395]]

to:

Confidence-based:
[[453, 47],
[120, 380]]

Therefore, false positives decreased from:

72→47

and the false-positive rate decreased from:

14.4%→9.4%

while accuracy increased from:

82.3%→83.3%

This shows that the confidence-based strategy successfully reduced false-positive object hallucinations while maintaining overall performance.

8. Category-Wise Analysis

The final confidence-based method was also evaluated separately on 300 samples from each POPE category.

Category	Accuracy	Precision	Recall	F1	FPR
Random	86.67%	95.83%	76.67%	85.19%	3.33%
Popular	85.33%	92.74%	76.67%	83.94%	6.00%
Adversarial	83.00%	87.79%	76.67%	81.85%	10.67%

The adversarial category was the most challenging.

It had the highest false-positive rate, showing that adversarial questions remain more difficult for the VLM.

However, the confidence-based approach still kept the FPR relatively controlled.

9. Task B Novelty

The main novelty in Task B is a **lightweight confidence-based decision calibration strategy**.

Rather than fine-tuning the complete VLM, the method uses the model's existing token probabilities for “yes” and “no” and applies a decision threshold.

This provides several advantages:

- No expensive VLM fine-tuning
- No additional training dataset required
- Simple to implement
- Computationally lightweight
- Directly targets false-positive predictions
- Allows the precision/recall trade-off to be controlled through the threshold

The experiment also demonstrates that naive prompting alone can worsen hallucination, making confidence-based calibration a more controlled alternative.

10. Limitations

Task A

The current implementation has several limitations:

1. The dataset was split into training and validation sets rather than using a completely independent test set.
2. The model was trained for only 20 epochs.
3. The implemented model performs semantic segmentation rather than dedicated instance segmentation.

4. Instance AP was therefore estimated using connected components.
5. Thin and low-contrast filaments remain difficult to detect.
6. Performance decreases at smaller image scales.

Task B

The main limitation is computational scope.

The POPE dataset contains 9,000 samples, but the main experiments were performed on a selected **1,000-sample subset**.

The category-wise experiment used **300 samples per category**.

Therefore, the reported results should be considered experimental evidence rather than a complete evaluation of the entire POPE dataset.

Another limitation is that the confidence-based approach changes the decision threshold but does not modify the underlying VLM.

11. Future Work

For Task A, future improvements could include:

- Training for more epochs
- Using stronger data augmentation
- Adding multi-scale training
- Using attention-based U-Net variants
- Using a dedicated instance-segmentation architecture
- Improving detection of thin filament structures
- Evaluating on a completely held-out test set

For Task B, future work could include:

- Evaluating all 9,000 POPE samples
 - Testing additional VLMs
 - Learning the confidence threshold automatically
 - Combining visual features with confidence calibration
 - Testing lightweight fine-tuning methods
 - Evaluating hallucination reduction on conversational VLM tasks
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12. Overall Conclusion

This project investigated two different computer vision challenges.

For **Task A**, a U-Net baseline was developed for solar filament segmentation and enhanced using an edge-aware Sobel loss. The edge-aware approach improved IoU from **0.4602 to 0.4671** and precision from **0.5930 to 0.6228**, although recall decreased slightly. The results

show that explicitly considering image boundaries can improve the quality of segmentation predictions, while thin and low-contrast filaments remain challenging.

For **Task B**, the BLIP VQA model was evaluated for object hallucination using the POPE benchmark. Naive prompting unexpectedly increased the false-positive rate from **14.40% to 51.30%**. A confidence-based decision strategy provided a better solution. At a threshold of 0.70, accuracy increased from **82.30% to 83.30%**, precision increased from **84.58% to 88.99%**, and FPR decreased from **14.40% to 9.40%**.

Overall, the experiments demonstrate two lightweight approaches: **edge-aware segmentation loss for improving structural boundaries** and **confidence-based calibration for reducing VLM object hallucinations**. Both approaches build upon existing models rather than requiring highly complex architectures or expensive training procedures.